Compact Lattice-Isometry Groups Survive the Benyamini Approximation

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Status. Partial affirmative result, likely valid, subject to human review.

1 Source question

Oikhberg and Tursi prove that, for every separable AM-space X and every $C > 1$, there are a Benyamini space X_C and a surjective lattice isomorphism $A : X \to X_C$ satisfying

$$\|x\| \le \|Ax\| \le C\|x\| \qquad (x \in X).$$

In Remark 2.4 they ask whether the construction can preserve the group of lattice isometries, or even a subgroup [1].

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that $y(t) = \lambda y(s)$ for any $y \in Y'$, then $\lambda = C^{n-m}$. Indeed, $y = Tx$ for some $x \in X$, so

$$\lambda = \frac{y(t)}{y(s)} = \frac{C^{1-m} x(\Psi(t))}{\psi(\Psi(t))} \cdot \frac{\psi(\Psi(s))}{C^{1-n} x(\Psi(s))} = C^{n-m} \cdot \frac{x(\Psi(t))}{x(\Psi(s))} \cdot \frac{\psi(\Psi(s))}{\psi(\Psi(t))}.$$

From this, it follows that $x(\Psi(t))/x(\Psi(s))$ is a constant on X. Either $\Psi(t) = \Psi(s)$, or $t' = \Psi(t)$ and $s' = \Psi(s)$ are "defining points" for $X \subset C(H)$ – that is, $x(t')/x(s')$ is independent of $x \in X$. Either way, $\lambda = C^{n-m}$.

Finally, we transform the sets $\widetilde{H}_n$ into sets K_n, whose points are separated by X'. By the preceding paragraph, if $t, s \in \widetilde{H}_n$ are such that $y(t) = \lambda y(s)$ for any $y \in Y$, then $\lambda = 1$. Define an equivalence relation on $\widetilde{H}$: $t \sim s$ if for all $y \in Y$, $y(s) = y(t)$. Clearly the equivalence classes are closed, hence each quotient space $K_n := \widetilde{H}_n/\sim$ is compact. Identify $\widetilde{H}/\sim$ with $K = (\cup_n K_n) \cup \{\infty\}$, which is the one-point compactification of $\cup_n K_n$. Define $\Phi : Y \to C_0(K)$ by setting, for $y \in Y$, $[\Phi y]([t]) = y(t)$, where $[t]$ is the equivalence class of t. Clearly Φ is a lattice isometry. $X' = \Phi(Y)$ is a Benyamini space, and $\Phi \circ T : X \to X'$ is a lattice isomorphism with desired properties. □

Remark 2.4. The Benyamini space X', constructed from X using Proposition 2.3, may have a different group of isometries. We do not know whether the Benyamini space can be constructed while preserving the group of isometries (or even a subgroup thereof).

Here "preserve" is made precise as the inclusion $AGA^{-1} \subseteq \mathrm{Iso}_{\mathrm{lat}}(X_C)$. The proof below gives this for every subgroup fixing a quasi-interior point and hence for every compact subgroup.

2 Result

Theorem 1 (Equivariant Benyamini approximation)**.** *Let X be a separable real AM-space, let $C > 1$, and let $G \le \mathrm{Iso}_{\mathrm{lat}}(X)$. Suppose that G fixes a quasi-interior point $e \in X_+$. Then Proposition 2.3 of [1] may be carried out so that its lattice isomorphism $A : X \to X_C$ satisfies*

$$\|x\| \le \|Ax\| \le C\|x\|, \qquad AGA^{-1} \subseteq \mathrm{Iso}_{\mathrm{lat}}(X_C).$$

Conjugation by A is also a topological group embedding for the strong operator topologies.

Corollary 2. *Every strong-operator compact subgroup of $\mathrm{Iso}_{\mathrm{lat}}(X)$ can be preserved in the sense of Theorem 1. In particular, every finite subgroup can be preserved. If $\mathrm{Iso}_{\mathrm{lat}}(X)$ itself is compact, the entire original isometry group survives as a subgroup of $\mathrm{Iso}_{\mathrm{lat}}(X_C)$.*

3 Proof

The fixed-point criterion

Use the canonical equivariant Kakutani representation. Namely, let

$$H = \{\varphi \in B_{X^*} : \varphi \text{ is a lattice homomorphism}\}$$

with its weak-star topology, and put $(Jx)(\varphi) = \varphi(x)$. The set H is compact, and the standard representation theorem for AM-spaces says that $J : X \to C(H)$ is a lattice isometry. Its common zero is $0 \in H$.

After scaling, assume $\|e\| \le 1$. If $\varphi \in H \setminus \{0\}$, then $\varphi(e) > 0$: otherwise φ vanishes on the principal ideal generated by e, hence on its dense closure X. Thus the function $\psi = Je$ vanishes precisely at 0.

For $n \ge 1$, form the compact level sets

$$H_n = \{\varphi \in H : C^{-n} \le \varphi(e) \le C^{1-n}\}.$$

Take disjoint copies $\widetilde{H}_n$, add the point at infinity when needed, and let $q : \widetilde{H} \to H$ forget the copy index and send infinity to 0. This is the level construction used in the proof of Proposition 2.3. Define, for $\widetilde{\varphi}_n \in \widetilde{H}_n$,

$$(Sx)(\widetilde{\varphi}_n) = \frac{C^{1-n}\varphi(x)}{\varphi(e)}, \qquad (Sx)(\infty) = 0.$$

The scalar multiplier lies in $[1, C]$. Since J is an isometry and every point of $H \setminus \{0\}$ occurs in a level,

$$\|x\| \le \|Sx\|_\infty \le C\|x\|.$$

Continuity at infinity follows as in the source proof: if the level indices tend to infinity, then $\varphi(e) \to 0$; quasi-interiority forces every weak-star cluster point of the corresponding φ's to be 0, hence $\varphi(x) \to 0$, while the displayed multiplier remains bounded by C.

For $g \in G$, define the homeomorphism

$$\alpha_g : H \to H, \qquad \alpha_g(\varphi) = \varphi \circ g.$$

Because g is a lattice isometry and $ge = e$, this map preserves every H_n. It therefore lifts to a homeomorphism $\widetilde{\alpha}_g$ of $\widetilde{H}$, fixing infinity. Directly from the definition,

$$S(gx) = Sx \circ \widetilde{\alpha}_g \qquad (x \in X). \tag{1}$$

Complete the source construction by identifying, within each level, points on which every member of $S(X)$ has the same value. Equation (1) shows that this equivalence relation is invariant under every $\widetilde{\alpha}_g$. Hence these homeomorphisms descend to the quotient compact pieces K_n. The source argument now gives a Benyamini space $X_C \subset C_0(K)$ and a lattice isomorphism $A : X \to X_C$ with the displayed distortion bounds. On X_C, the conjugate AgA^{-1} is composition with the descended homeomorphism, and therefore is a surjective lattice isometry. This proves the algebraic assertion. Conjugation by the fixed bounded isomorphisms A and A^{-1} is continuous in both directions for point-norm convergence, proving the strong-operator statement.

Compact groups supply the fixed point

Let $G \leq \mathrm{Iso}_{\mathrm{lat}}(X)$ be compact in the strong operator topology. Since X is separable, choose a quasi-interior point $e_0 \in X_+$. With m normalized Haar measure on G, the Bochner integral

$$e = \int_G g e_0 \, dm(g)$$

exists, is positive, and is fixed by G. It remains to check that e is quasi-interior.

Let $0 \neq f \in X_+^*$. Quasi-interiority of e_0 gives $f(e_0) > 0$. The function $g \mapsto f(ge_0)$ is continuous and nonnegative. If its Haar integral were zero, it would vanish on all of G, contradicting its positive value at the identity. Hence $f(e) > 0$ for every nonzero $f \in X_+^*$. This strict positivity characterizes quasi-interior points: otherwise the proper closed ideal generated by e would admit a nonzero positive functional on the quotient that annihilates e. Thus e is quasi-interior, and Theorem 1 applies.

4 Scope and novelty

The theorem supplies a faithful conjugate copy of G in the new isometry group. It does not assert that no additional lattice isometries appear. It also leaves open arbitrary noncompact subgroups that have no fixed quasi-interior point.

A bounded search on 9 August 2026 covered the run indexes, the parsed arXiv corpus, exact-phrase searches for Remark 2.4, and close searches combining Benyamini spaces, compact groups, and quasi-interior points. It found the source and the authors' later $C_0(L)$ group-display paper [2], which explicitly asks for a general AM-space extension, but no occurrence of the equivariant criterion or the compact-subgroup corollary above. Novelty confidence is moderate pending expert review.

References

[1] T. Oikhberg and M. A. Tursi, *Renorming AM-spaces*, Proc. Amer. Math. Soc. **150** (2022), 1127–1139; arXiv:2011.04140.

[2] T. Oikhberg and M. A. Tursi, *Lattice renormings of $C_0(X)$ spaces*, Banach J. Math. Anal. **18** (2024), article 76; arXiv:2405.11148.